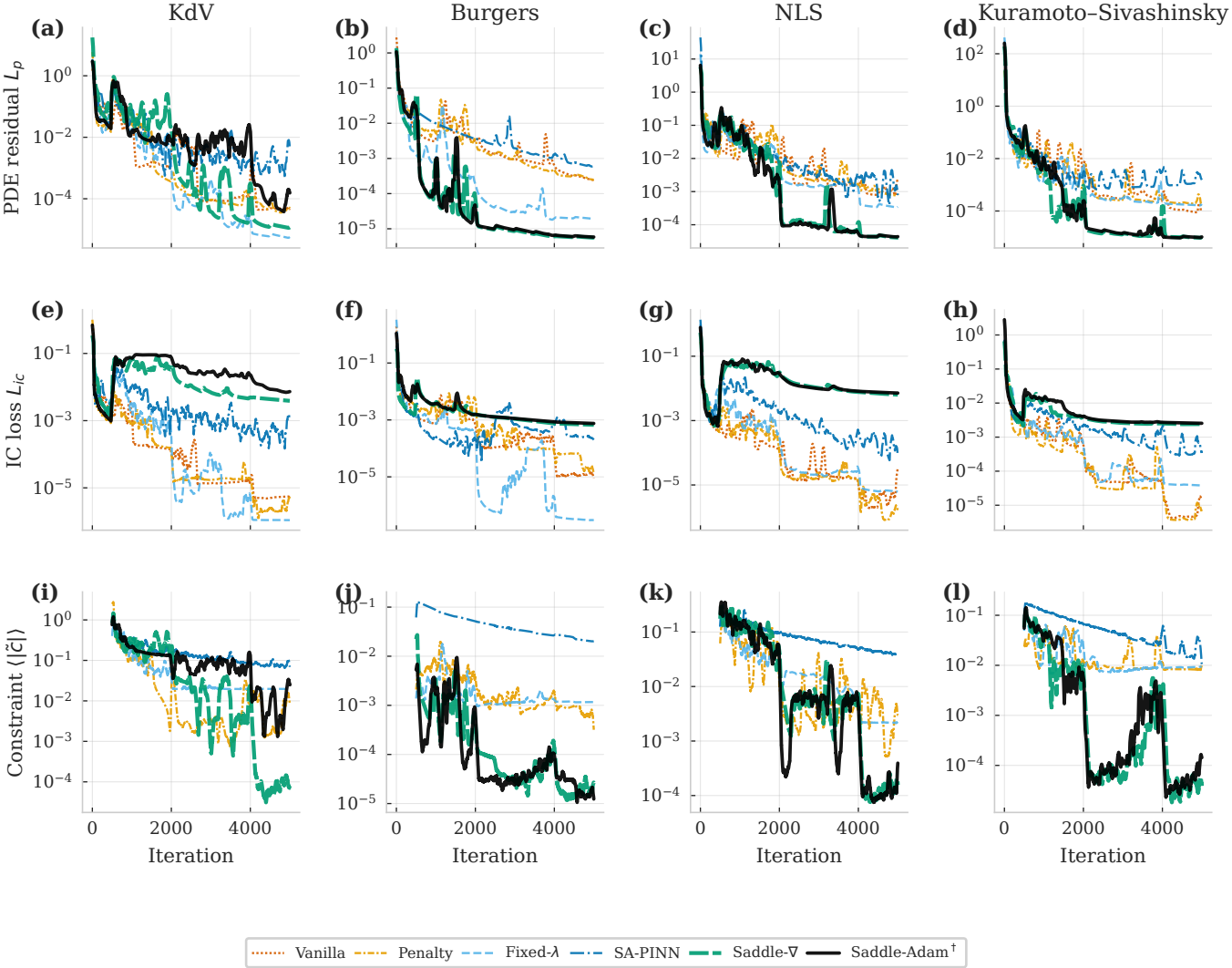
KdV
Burgers
NLS
Kuramoto–Sivashinsky
(a)
(b)
(c)
(d)
(e)
(f)
(g)
(h)
(i)
(j)
(k)
(l)
PDE residual L_p
IC loss L_{ic}
Constraint ($|\tilde{c}|$)
Iteration
Vanilla
Penalty
Fixed-λ
SA-PINN
Saddle-∇
Saddle-Adam†